Logbook: Efficient ResNet for CIFAR-10 (5M parameter limit)

This logbook records all the approaches that we tried and all the results that we got. The language follows ASD-STE100 Simplified Technical English. Sentences are short. Each sentence gives one idea.

1. Task and rules

The task is a Kaggle competition for a course. The rules are:

1. Train a modified ResNet. The network must have skip connections.
2. The model must have a maximum of 5,000,000 trainable parameters.
3. Train from zero. Do not use pre-trained weights.
4. Use only the CIFAR-10 training data. Do not use other datasets.
5. Do not label the test images by hand. Do not train on the test images.
6. Optimizers, augmentation, regularization, schedules and distillation are permitted.
7. The competition closes on 2026-11-06 at 23:59. The report is due on 2026-11-09.

The metric is classification accuracy. Kaggle shows a public score on half of the hidden images. The private score on the other half sets the final rank.

2. Data

The `data/` folder contains these files:

File	Contents
`train_labels.csv`	50,000 rows: `id`, `filename`, `label` (0 to 9)
`train_images/`	50,000 RGB PNG images, 32 × 32 pixels
`test.csv`	2,000 rows: `id`, `filename`
`test_images/`	2,000 RGB PNG images, 32 × 32 pixels
`sample_submission.csv`	2,000 rows: `id`, `label`

Each class has exactly 5,000 training images. The test IDs are 0 to 1999. They match the sample submission.

`src/prepare_data.py` decodes all PNG files one time. It writes `data/cifar10_cache.npz`. The channel means of the training images are 0.4914, 0.4822 and 0.4465. These are the standard CIFAR-10 values.

The test set has 2,000 images. This is the same size as CIFAR-10.1 (Recht et al., 2018). CIFAR-10.1 is a new test set that has a small distribution shift from CIFAR-10. Models usually get 4 to

10 points less accuracy on CIFAR-10.1 than on CIFAR-10. Our Kaggle scores agree with this. Thus, we think that the hidden set is similar to CIFAR-10.1.

3. Method for evaluation

We did not select models with the public Kaggle score. The public split has only about 1,000 images. One score has a standard error of about 1.1 points. Selection on that split can overfit.

We use a holdout set from the training data. The holdout has 5,000 images: 500 from each class. `src/train.py` makes the split with a fixed seed (`--val 5000 --seed 0`). The model trains on the other 45,000 images.

All holdout accuracies in this logbook use test-time augmentation (TTA), unless the text says otherwise. TTA averages the outputs for the image, its mirror, and two 1-pixel shifts.

For the final submission, we train the selected recipe again on all 50,000 images (`--val 0`).

Rule for a change: we keep a change only if it adds at least 0.3 points on the holdout.

4. Hardware

Hardware	Use	Speed
Apple M4 Pro, MPS backend	Phase 1 and 2	About 23 s per epoch for a 4.8M AirResNet
NVIDIA A100 40 GB, NYU HPC	Phase 3 to 6	About 6.6 s per epoch for the 4.7M PreActResNet

All data stays on the GPU. The GPU does the crop, flip and cutout operations. TrivialAugment runs on CPU worker processes. The workers prepare the next epoch during training.

5. Literature research

We did two literature surveys. We used three research skill packages together:

- `academic-research-plugin` (literature-survey): search on arXiv, themes, gaps, proposals, BibTeX.
- `academic-research-skills` (deep-research): WHY / HOW / WHAT notes, method weaknesses, a devil's-advocate check.
- `scientific-agent-skills` (literature-review): search log, screening, citation check.

The full reports are in `output/literature-survey/` and in `docs/RESEARCH_REPORT.md`.

Main results from the research:

1. Accuracy on a shifted test set follows accuracy on the original test set on a straight line. For CIFAR-10.1: new accuracy = 1.62 × CIFAR-10 accuracy – 65.51 (Recht et al., 2018). Thus, each +1 point on the holdout gives about +1.6 points on the hidden set.

2. Almost no method moves a model above this line without more training data (Taori et al., 2020).
3. Cutout and wider networks help more on CIFAR-10.1 than on CIFAR-10.
4. Test-time adaptation (BN adaptation, TENT) uses the test images. The rules do not permit it. It also makes CIFAR-10.1 accuracy worse (Zhao et al., 2023).

6. Experiments in time order

Phase 1: AirResNet on the M4 Pro (2026-10-29)

The first model is "AirResNet". It comes from airbench96 (Jordan, 2024). It has a frozen 2×2 whitening layer, three stages of conv → max-pool → BN → GELU, and residual pairs of 3×3 convs. The recipe uses Nesterov SGD, label smoothing 0.2, flip, 4-pixel translate, 12-pixel cutout, and a Lookahead EMA.

We tested three width and depth settings for 40 epochs:

Run	Widths	Residual pairs	Params	Holdout
`w128-256-352_d111_e40`	128/256/352	1/1/1	4.84M	**95.58%**
`w96-256-352_d211_e40`	96/256/352	2/1/1	4.80M	95.36%
`w96-192-384_d111_e40`	96/192/384	1/1/1	4.34M	95.10%

The widest model was best. Its Kaggle public score was **0.850**.

Phase 2: longer training and TrivialAugment (2026-10-29)

Run	Epochs	TrivialAugment	Holdout	Kaggle
`w128-256-352_d111_e40`	40	no	95.58%	0.850
`w128-256-352_ta_e40`	40	yes	94.98%	
`w128-256-352_e120`	120	no	96.02%	
`w128-256-352_ta_e120`	120	yes	**96.20%**	0.860

Longer training added 0.4 to 0.6 points. TrivialAugment decreased accuracy at 40 epochs. It added 0.2 points at 120 epochs.

Diagnostic: we made the holdout images brighter and with more contrast. This matched the pixel statistics of the test images. The accuracy decreased by a maximum of 0.4 points (`src/shift_check.py`). Thus, brightness and contrast do not cause the Kaggle gap.

Phase 3: rotation self-supervision (SLA) on the A100 (2026-10-29)

SLA (Lee et al., 2020) trains the network on the 4 rotations of each image. The labels are joint labels (class × rotation), 40 classes. A second head learns from the averaged rotation prediction (self-distillation). At test time, we use the normal head, the 4-rotation vote, or the mean of the two. We checked our code against the paper and the official code (hankook/SLA).

Run	Seed	Holdout (head / vote / both)	Control without SLA	Kaggle
`sla_ta_e120_b1024`	0	96.30% / 96.32% / **96.40%**	95.90% (`ta_e120_b1024_a100`)	0.863
`sla_ta_e120_b1024_s1`	1	**96.68%** / 96.30% / –	96.28% (`ta_e120_b1024_s1`)	0.857 (control)

SLA added 0.4 to 0.5 points on two seeds. But all AirResNet models stayed near 0.86 on Kaggle.

Phase 4: standard pre-activation ResNet (2026-10-29)

All AirResNet models were about 4 points below the CIFAR-10.1 line. Thus, we changed the architecture family.

The new model is a standard pre-activation ResNet (He et al., 2016b):

- A 3×3 stem conv with 64 channels.
- Three stages with 4, 4 and 3 basic blocks. The widths are 64, 128 and 256.
- Each block is BN → ReLU → conv 3×3 → BN → ReLU → conv 3×3, plus a skip connection.
- The skip is the identity, or a 1×1 conv when the shape changes.
- Global average pooling and one linear layer.

The recipe: Nesterov SGD, lr 0.4 at batch 512, weight decay 5e-4, 5 warmup epochs, cosine schedule, label smoothing 0.1, TrivialAugment, flip, 4-pixel translate, 12-pixel cutout, 200 epochs.

Run	Params	Holdout	Kaggle
`preact_ta_e200`	4,696,394	96.82%	**0.884**

The holdout gain was 0.4 points. The Kaggle gain was 2.1 points. The architecture family has an effect on the shift.

Phase 5: PreActResNet + SLA and the final model (2026-10-29 and 2026-10-30)

Run	Data	Holdout (head / vote / both)	Kaggle
`preact_sla_ta_e200`	45,000	**97.56%** / 97.42% / 97.54%	
`preact_sla_ta_e200_full`	50,000	–	**0.909**

SLA added 0.74 points on the PreActResNet. This is our best recipe. A second seed (`preact_sla_ta_e200_s1`, seed 1, its own holdout split) got **97.78%** (head) and 97.54% (vote). Thus, the recipe gives 97.56% to 97.78% on the holdout. The difference between seeds is about 0.2 points. We trained it again on all 50,000 images (92 minutes on the A100). This is the final `submission.csv`. The prediction is the mean of the softmax of the normal head and the softmax of the 4-rotation vote.

Phase 6: other methods from the research report (2026-10-30)

We tested each method against the correct baseline. No method beat the final recipe.

Run	Method	Holdout	Baseline	Result
`preact_ta_e200_sam05`	SAM, ρ = 0.05 (Foret et al., 2021)	97.18%	96.82%	+0.36
`preact_sla_ta_e200_sam05`	SAM + SLA	97.58% (both)	97.54% (both)	+0.04, not kept (2× cost)
`papa4_preact_ta_e200`	PAPA, 4 copies (Jolicoeur-Martineau et al., 2024)	97.18–97.20% per copy	96.82%	below 97.56%
`shake2x40d_ta_e600`	Shake-Shake-26 2x40d, 600 epochs (Gastaldi, 2017)	97.16%	97.56%	below
`preact_blur_ta_e200`	Anti-aliased downsampling (Zhang, 2019)	97.00%	96.82%	+0.18, not kept
`preact_ta_e400`	400 epochs	97.12%	96.82%	+0.30, below 97.56%
`preact_kd_sla2_ta_e200`	Distillation from 2 SLA teachers (Hinton et al., 2015)	97.18%	97.56%	below

Notes:

- SAM helps the plain ResNet. SAM does not help on top of SLA.
- In the PAPA run, the averaged model got only 96.20%. The cause was a bug: the code calculated the BN statistics again on clean images. With BN statistics from augmented images, the averaged model got 96.94%. We changed the code. The averaged model now uses the mean of the BN statistics of the copies.
- Shake-Shake needs a long schedule. We trained it for 600 epochs over two HPC sessions with checkpoints (`--resume`).

Phase 7: CutMix inside SLA (2026-10-31)

`preact_sla_cutmix_e200`: the final recipe with CutMix (Yun et al., 2019). With probability 0.5, each batch gets one box from a shuffled partner image. The box goes in before the four rotations, and the losses are mixed by area. The run also kept a weight average (EMA, decay 0.999).

Prediction	CutMix run	Final recipe (seed 0)
Normal head	97.42%	97.56%

Prediction	CutMix run	Final recipe (seed 0)
4-rotation vote	97.52%	97.42%
Weight average, normal head	96.98%	–

CutMix did not add accuracy. The weight average was 0.4 points worse than the live network at the end of the cosine schedule. We queued a 300-epoch CutMix run for the next session.

`preact_sla_cutmix_e300`: the same CutMix run for 300 epochs (seed 0), without the weight average. Training time was 124 minutes on the A100.

Prediction	CutMix, 300 epochs	Final recipe (seed 0, 200 epochs)
Normal head	97.66%	97.56%
4-rotation vote	97.64%	97.42%
Head + vote	97.70%	–

The gain is +0.10 points. This is less than our limit of +0.3 points (the seed spread is approximately 0.2 points). Thus, we did not change the final model. We saved the predictions in `submissions/submission_preact_sla_cutmix_e300.csv`. We did the 300-epoch run without CutMix in the next HPC session (see below).

Phase 8: Per-copy augmentation and 300 epochs (2026-10-31)

We did two more runs on the A100 with seed 0:

- `preact_sla_percopy_e200`: each of the four rotated copies of an image gets its own random augmentation.
- `preact_sla_ta_e300`: the final recipe for 300 epochs, without CutMix. This run is the control for the CutMix run.

Prediction	Per-copy, 200 epochs	300 epochs	CutMix, 300 epochs	Final recipe, 200 epochs
Normal head	97.58%	97.68%	97.66%	97.56%
4-rotation vote	97.52%	97.68%	97.64%	97.42%
Head + vote	97.66%	97.74%	97.70%	97.54%

Per-copy augmentation did not change the accuracy. Thus, we did not do the two-view run. The 300-epoch runs got approximately +0.1 points, with or without CutMix. Thus, CutMix does not add accuracy. All gains are less than our limit of +0.3 points. We did not change the final model. We saved the predictions of the two runs in `submissions/`.

7. Distillation result

The student is a PreActResNet without SLA. The teachers are `preact_sla_ta_e200` and `preact_sla_ta_e200_sam05`. Both teachers trained on the same 45,000 images. Thus, the holdout stays clean. The teacher and the student see the same augmented image (Beyer et al., 2022). The loss is 0.5 × cross-entropy + 0.5 × KL divergence at temperature 4.

Result: the student got **97.18%** on the holdout. This is +0.36 points more than the same ResNet without distillation (96.82%). But it is 0.38 points less than the teachers' recipe (PreActResNet + SLA, 97.56%). Thus, we did not keep it.

8. Kaggle public scores

Submission	Model	Holdout	Kaggle public
1	AirResNet, 40 epochs	95.58%	0.850
2	AirResNet + TrivialAugment, 120 epochs	96.20%	0.860
3	AirResNet + SLA, 120 epochs	96.40%	0.863
4	AirResNet, seed 1, 120 epochs	96.28%	0.857
5	PreActResNet, 200 epochs	96.82%	0.884
6	PreActResNet + SLA, all 50k (final)	97.56% (45k version)	**0.909**
7	PreActResNet + SLA + per-copy augmentation, 45k	97.58%	0.896
8	PreActResNet + SLA + CutMix, 300 epochs, 45k	97.66%	0.902
9	PreActResNet + SLA, 300 epochs, 45k	97.68%	0.904
10	PreActResNet + SLA, 200 epochs, 45k (control for 7 to 9)	97.56%	0.903

Submissions 7 to 10 trained on 45,000 images. On the same 45k data, the changes are in the range –0.007 to +0.001 public points. The final model trained on all 50,000 images and still has the best public score. We did not use these public scores to select a model.

9. Final model

Item	Value
Architecture	Pre-activation ResNet, stages 4/4/3 blocks, widths 64/128/256, average pooling
Heads	10-class head + 40-class joint head (class × rotation)
Trainable parameters (torchsummary, forward pass)	4,696,394
Trainable parameters (all, with the 40-class head)	4,706,674
Training data	All 50,000 CIFAR-10 training images

Item	Value
Training	200 epochs, batch 512, Nesterov SGD, cosine schedule, SLA+SD, TrivialAugment, cutout
Test-time augmentation	Mirror and 1-pixel shifts; normal head + 4-rotation vote
Holdout accuracy (same recipe on 45,000 images)	97.56%
Kaggle public accuracy	0.909

10. Lessons

1. Do not select models with the public leaderboard. Use a holdout from the training data.
2. The architecture family has an effect on accuracy under shift. A fast-training airbench net was 1.4 points worse on Kaggle than a standard ResNet with the same holdout accuracy.
3. SLA (rotation labels) gave the largest gain for its cost: +0.4 to +0.7 points.
4. SAM, PAPA, distillation, longer schedules and anti-aliasing each gave +0.2 to +0.4 points on the plain ResNet. None of them beat SLA.
5. Calculate BN statistics on data that is similar to the training data. Clean-image statistics decreased the PAPA model by 1 point.
6. Make long runs resumable. HPC jobs stop after 3 to 5 hours.

11. Pipeline audit (2026-10-31)

We did a full review of the code, the data and our method. We also read more papers about the shift to CIFAR-10.1. The review did not start a training run. It did not change the data or the final submission.

11.1 Result

We did not find an error that causes the gap between the holdout (97.6%) and Kaggle (0.90). These items are correct:

- All 50,000 training images and all 2,000 test images are 32 × 32 RGB PNG files. The cache is equal to the files.
- The labels and the IDs are equal to the CSV files. No test image is an exact copy of a training image.
- The training images and the test images get the same normalization.
- The SLA joint label is `class × 4 + rotation` during training and during prediction. The rotation order agrees.
- Augmentation occurs before the rotation. BN and bias parameters get no weight decay.
- The final model has 4,706,674 parameters and all 200 epochs completed.
- All 15 submission files have 2,000 rows, the correct ID order and labels 0 to 9.
- No code trains on the test images, labels them or adapts to them.

The test images are a little brighter and have more contrast than the training images:

Pixel statistic	Training	Test
Channel mean (R, G, B)	0.491, 0.482, 0.447	0.513, 0.498, 0.461
Channel std (R, G, B)	0.247, 0.243, 0.262	0.265, 0.262, 0.275
Spread of image brightness	0.125	0.144

`src/shift_check.py` shows that this difference costs a maximum of 0.4 points. Thus, it does not cause a 7-point gap. The line of Recht et al. (2018) gives approximately 92.8% at 97.7% holdout accuracy. Miller et al. (2021) show the same line for 1,060 models. Our 0.909 is approximately 1.6 points below the line. The public split has approximately 1,000 images. One standard error is approximately 0.9 points. Thus, we think that the gap comes from the shift in the hidden images, and not from an error.

11.2 Problems that we found

Problem	Location	Effect
We compared runs with the normal head. But we submit the normal head + the 4-rotation vote.	`docs/LOGBOOK.md`, Phase 8	SLA-300 gets +0.20 points on head + vote (97.54% → 97.74%), not +0.12.
The seed also selects the holdout split. Seed 0 and seed 1 have only 527 holdout images in common.	`src/train.py:105`	The seed spread of 0.2 points is not a correct measurement. We cannot test a seed-0 + seed-1 ensemble on either holdout.
The weight average (EMA) calculates BN statistics on upright images only. Training used 4 rotations and CutMix.	`src/train.py:272`, `src/train.py:428`	The EMA result (−0.4 points) can be too low.
SAM does the second forward pass in training mode. Thus, the BN statistics change two times in each step.	`src/train.py:539–552`	The SAM result can be too low.
`make_submission.py` uses the normal head only. `export_preds.py` uses head + vote.	`src/make_submission.py:20`	The final CSV is correct (it came from `export_preds.py`). Use `export_preds.py` for all SLA submissions.
A resumed run does not keep the TrivialAugment epoch or the random state.	`src/train.py:161`, `src/train.py:450`	Small. A resumed run is not equal to a run without a stop.
CutMix uses the NumPy random generator without a seed.	`src/train.py:262`	No effect on accuracy. Runs are not reproducible.

The weight of the normal head and the vote is 0.5 each. We calculated other weights from the saved holdout outputs:

Run	Vote only	0.25 head	0.5 head (submitted)	0.75 head	Head only
SLA, 200 epochs	97.42%	97.50%	97.54%	97.64%	97.56%

Run	Vote only	0.25 head	0.5 head (submitted)	0.75 head	Head only
SLA, 300 epochs	97.68%	97.84%	97.74%	97.70%	97.68%
SLA + CutMix, 300 epochs	97.64%	97.70%	97.70%	97.66%	97.66%

The best weight is different for each run. Thus, a part of this gain is noise.

11.3 Decisions

1. The 50k SLA-300 retrain is the best next submission. We export it with `scripts/export_preds.py`.
2. We do not use an ensemble of different networks. Two networks have 9.4M parameters. This is more than the 5M limit.
3. We do not adapt BN to the test images. The rules do not permit it. It also increased the CIFAR-10.1 error from 12.8% to 19.0% (Zhao et al., 2023).
4. Other tests from the audit are optional: one fixed head/vote weight, a BN fix for EMA, symmetric 1-pixel TTA, and a wider ResNet with 80/160/320 channels.

The 0.3-point rule is a limit for cost. It is not a statistical test. On 5,000 images, it finds a true +0.3-point gain only 30% to 55% of the time. For the next comparisons, we will also give the number of images that each change fixes and breaks (paired McNemar test).